7 The decay of radioactive materials is a *random* process. On average, nuclides which decay quickly exist only for a short period of time, while nuclides which decay slowly last longer. One difficulty that arises with these calculations is when the radioactive material is a mixture of two or more nuclides. This question considers the case when a mixture of two radioactive nuclides is present. In decommissioning a nuclear power station, this difficulty is compounded by the presence of about a hundred different radioactive nuclides in significant quantities.

(a) Explain what it means to say that radioactive decay is a *random* process.

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.....

.....

..... [2]

(b) State two physical quantities which do not cause a change in the rate of decay of a radioactive material.

1. .....

2. ..... [2]

(c) Fig. 7.1 gives the variation with time of the total activity A_{mix} of a mixture of cobalt and nickel together with the separate activities A_c and A_N due to cobalt and nickel.

time/year	A_c / Bq	A_N / Bq	A_{mix} / Bq	ln (A_{mix} / Bq)
0	6900	250	7150	8.87
5	3540	241	3781	8.24
10	1820	232	2052	7.63
20	479	215	694	6.54
30	126	199	325	5.78
40	33.3	185	218	5.39
50	8.79	172	181	5.20
60	2.32	159	161	5.08
70	0.611	147	148	4.99
80	0.161	137	137	4.92
90	0.425	127	127	4.85
100	0.0112	118	118	4.77

Fig. 7.1

Fig. 7.2 shows the variation of ln A_{mix} with respect to time.

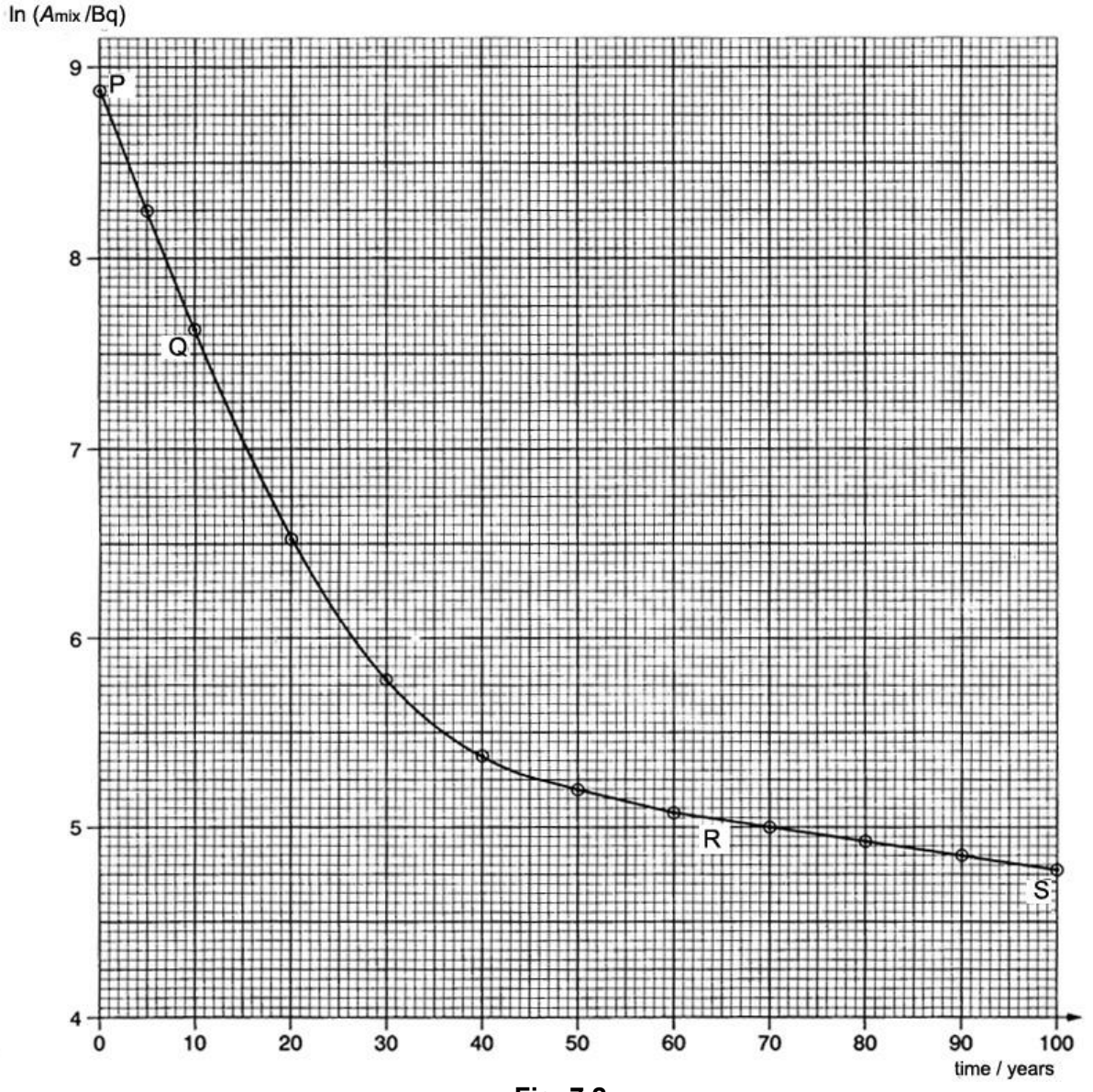

Fig. 7.2

(i) Explain the following:

1. PQ on the graph corresponds mainly to the decay of cobalt.

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.....

..... [2]

2. RS on the graph corresponds mainly to the decay of nickel.

.....

.....

..... [1]

3. The shape of QR is a curve.

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.....

.....

..... [2]

(ii) Determine the gradients of

1. PQ,

2. RS.

gradient of PQ = ... $year^{-1}$

gradient of RS = ... $year^{-1}$ [4]

(iii) Given that the general decay law is of the form $x = x_o \, exp\,(-\lambda\, t)$, use the gradients found in **(ii)** to estimate values of the decay constants for the cobalt and the nickel nuclides.

decay constant of cobalt = ... $year^{-1}$

decay constant of nickel = ... $year^{-1}$ [1]

(iv) Use your answer to **(iii)** to calculate the half-lives of the cobalt and nickel.

half-life of cobalt = ... years

half-life of nickel = ... years [3]

(d) In an actual reactor, activities of radioactive materials can often be 10^{12} times larger than those given in Fig. 7.1. Explain when and why each of these two nuclides would pose the greater hazard.

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..... [3]